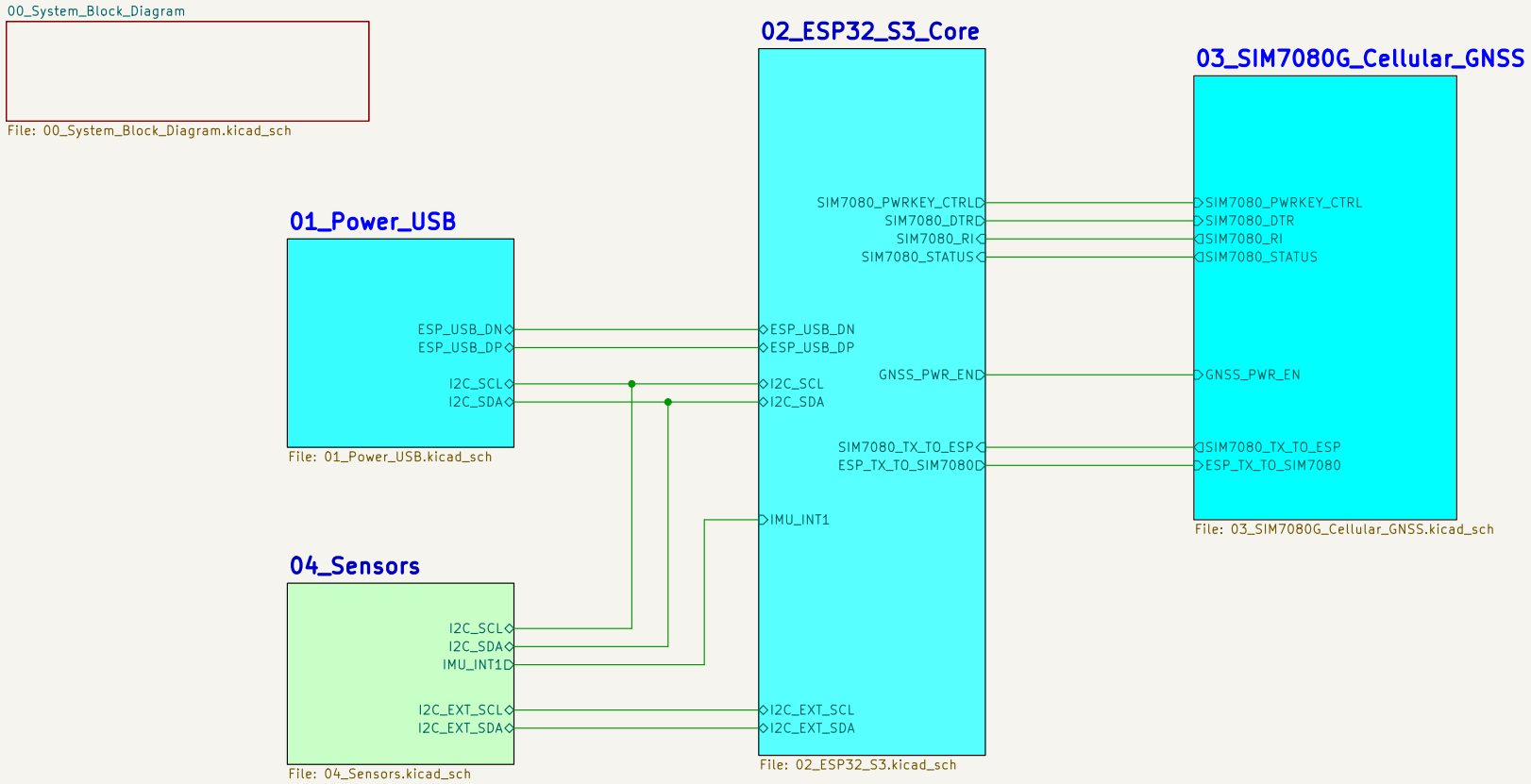
EdgeLink S3 Cellular — System Interconnect

Hierarchical Sheet Interconnections & Signal Routing



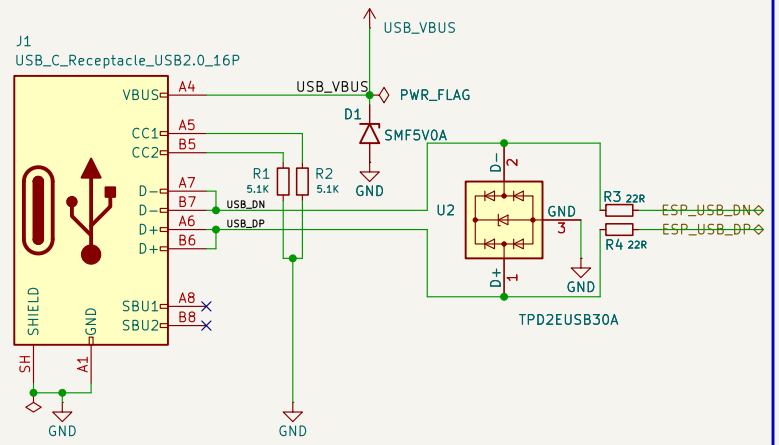
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KiCad E.D.A. 10.0.6		Id: 1/6

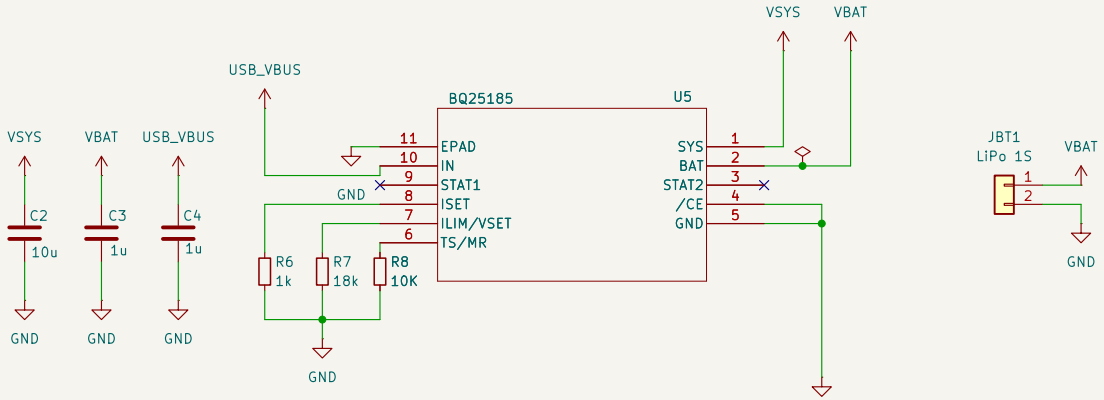
Power Input, Charging & Regulation

USB-C Power & Data



Hardware Testing : Pending
Design Review : Required

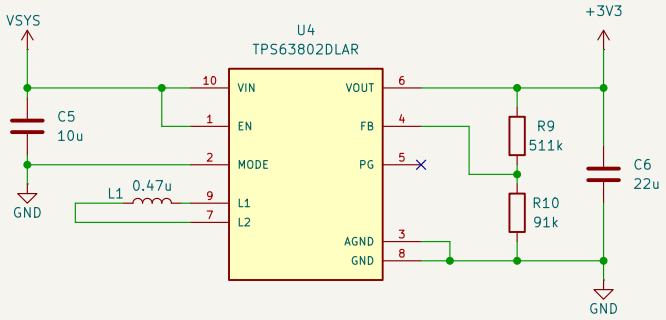
Battery Charger & Power Path



Charge Current Setting
R6 = 1 kΩ sets BQ25185 charge current to approximately 300 mA.
Selected as a practical default for a small single-cell Li-ion/LiPo battery.
Adjust R6 if a different battery capacity or charge-current limit is required.

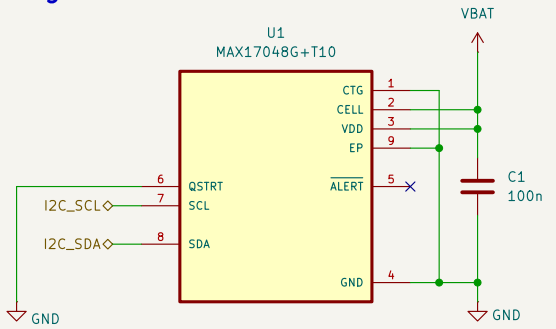
Hardware Testing : Pending
Design Review : Required

3.3 V Buck-Boost Regulation



Hardware Testing : Pending
Design Review : Required

Battery Monitoring

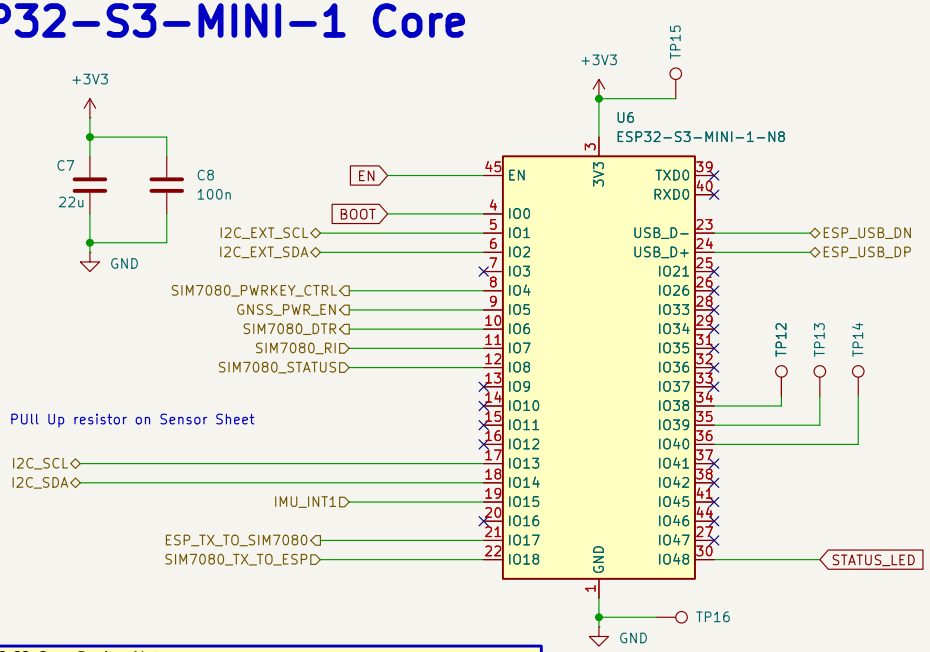


Hardware Testing : Pending
Design Review : Required

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KiCad E.D.A. 10.0.6		Id: 2/6

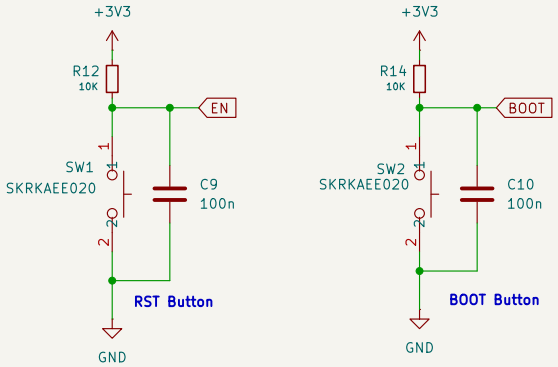
ESP32-S3-MINI-1 Core



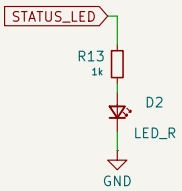
ESP32-S3 Core Design Notes

- ESP32-S3-MINI-1-N8: Wi-Fi and Bluetooth LE.
- Module supply: +3V3, with local decoupling.
- USB D- / D+: GPIO19 / GPIO20.
- EN and BOOT: reset and download-mode buttons.
- Internal I2C: GPIO13 / GPIO14, fuel gauge and sensors.
- External I2C: GPIO1 / GPIO2, separate signal pair.
- BMI323 INT1: GPIO15. Status LED: GPIO48.
- Modem UART: GPIO17 / GPIO18, via U7 level shifter.
- Keep the module antenna region unobstructed.
- Rev A hardware testing and firmware remain pending.

Reset & Boot Control



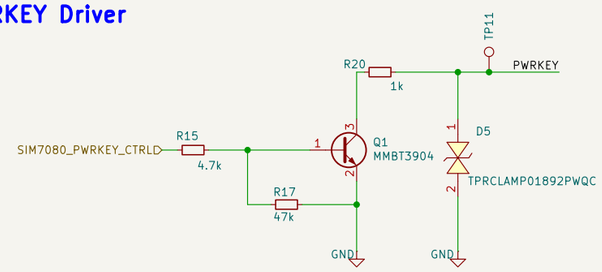
User Status LED



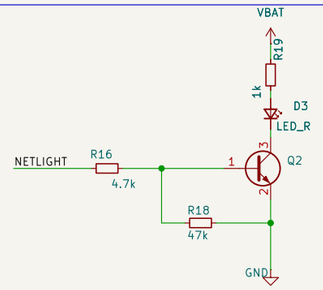
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Title:	ESP32-S3 Wireless Controller Core	
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 3/6

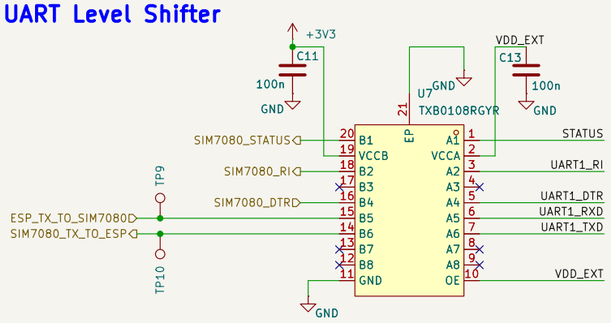
PWRKEY Driver



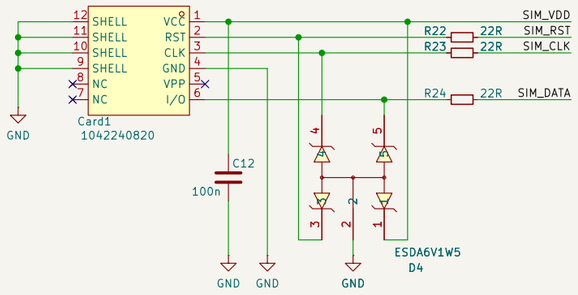
NETLIGHT



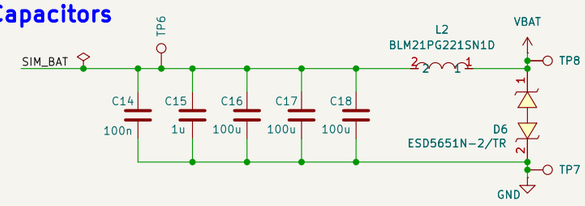
UART Level Shifter



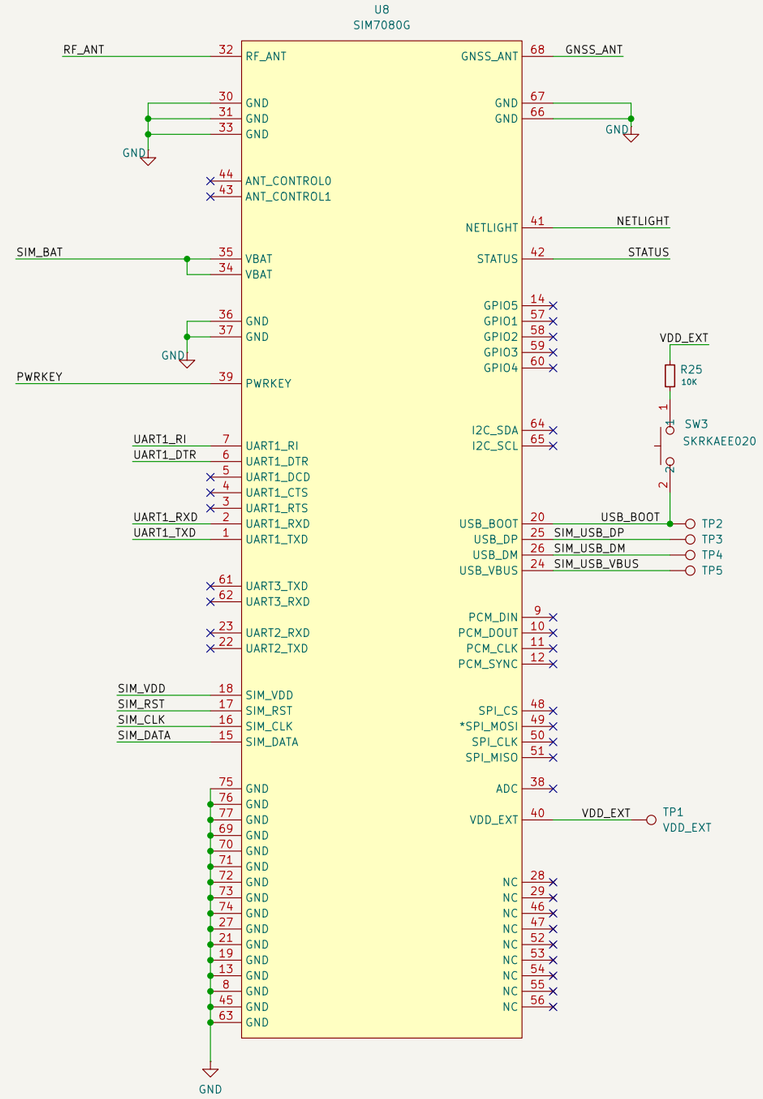
SIM Card Interface



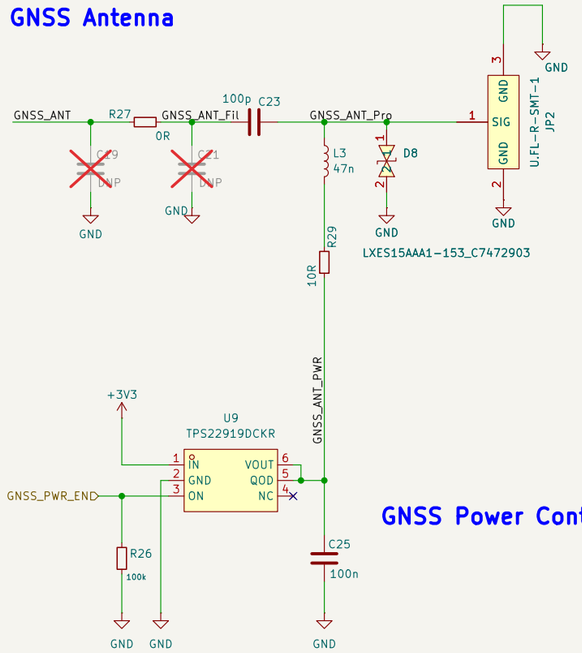
Decouple Capacitors



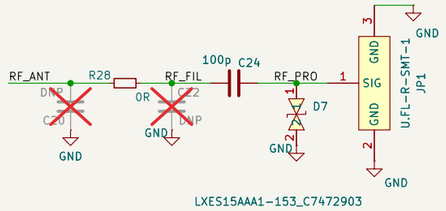
SIM7080G Modem



GNSS Antenna



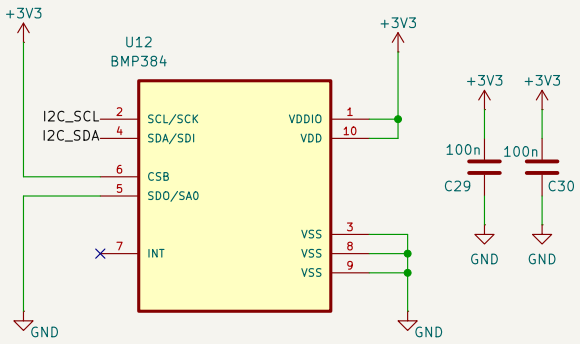
LTE – NB IoT Antenna



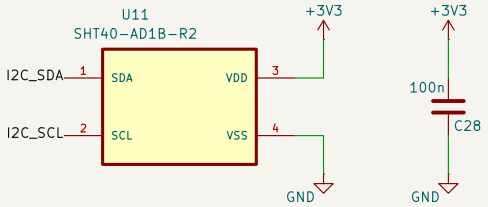
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04 — Motion & Environmental Sensing

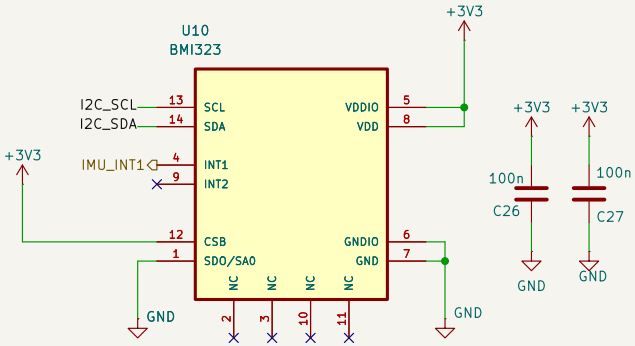
BMP384: barometric pressure



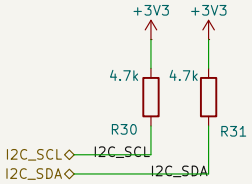
SHT40: temp + relative humidity



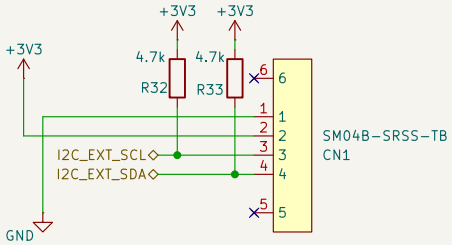
BMI323: 6-axis motion



I2C Pullups



External I2C Breakout



Product options:
- BMI323: 6-axis motion / wake-on-motion / tamper sensing
- SHT40: temperature + relative humidity
- BMP384: barometric pressure
- Shared 3.3 V I2C bus with one 4.7 kΩ pull-up pair
- SHT40 to be placed at PCB edge away from modem, ESP32 and power converters.

Sheet: /04_Sensors/		
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KiCad E.D.A. 10.0.6		Id: 5/6

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